

H-LAYER analytic closure (v1.1): the T-016 to T-021 arc, the lattice-off-shell

TECT verification-first repository · claim B2-PROPA-HLAYER

first issued 2026-06-10 · this version issued 2026-06-10 · v1.1

1. Purpose and scope

After the SC-SCOPE lift, B1-RH-ENUM and B2-PROPA-HLAYER rest on the sole hypothesis H-LAYER. The T-016 to T-021 arc closes H-LAYER's analytic residuals (STEP-5B) class-wide on the registered admissible class. This note consolidates the arc and – per the operator adversarial review – records the class-specificity of the $T' \leq 13$ pin and the universal fallback. Scope: operating endpoint $I = 2 \times 10^{-3}$, ROBUSTNESS-MU2 [$\times 0.5, \times 2$].

2. The H-LAYER decomposition

The isotropic layer is the infimum iff no admissible competitor lowers F : the comparison splits into (D) the diagonal sector (Math427/H-diag), (O) the off-diagonal Bloch sector (RES-1), and (S) the STEP-5B selection floor + SC-SCOPE third cumulant.

3. The arc (T-016 to T-021)

T-016 Diagonal isotropy: STRICT curvature-protected minimum; Hessian block-diagonal in angular momentum ($\Phi''_{\text{diag}} = \frac{3}{2}u_{\text{eff}} = 4.03 > 0$ at $l = 0$; entropy $\geq \frac{1}{2}r_R^2 = 4.64 \times 10^{-2} > 0$ at $l \geq 1$, analytic).

T-017 Selection floor: weighted Lemma A gives $K_{\text{floor}}(c) \leq T'(Q)$ for any amplitudes ($\chi(P)$ bypassed); on the registered class $T' \leq 13$.

T-018,T-019 Off-diagonal: B^{cond} diagonal, $R_{\text{lead}} \leq 0.650$ (registered class); NO A-independent attractive Fock exchange (local functional, $+\frac{3}{2}u_{\text{eff}}$ stabilising).

T-020 Off-diagonal: band positive iff $\rho R_{\text{lead}} < 1$; at $\rho = 1$, $R_{\text{lead}} \leq 0.650 < 1$ – second-cumulant stability class-wide.

T-021 SC-SCOPE: $\text{joint}(\rho_{\text{lat}}) > 1 \iff T' < 60.4$; on the registered class $T' \leq 13 < 60.4$, $\text{joint} \geq \times 1.095$.

4. Final analytic status (registered class)

(D) diagonal isotropy	STRICT minimum, analytic (T-016)
(O) off-diag A-independent	stabilising $+ \frac{3}{2}u_{\text{eff}}$ (T-019)
(O) off-diag A-dependent, 2nd cum.	$\rho R_{\text{lead}} \leq 0.650 < 1$ (T-018,020)
(S) floor $K_{\text{floor}} \leq T' \leq 13$	registered class (T-017)
(S) SC-SCOPE 3rd cumulant	joint $\geq \times 1.095$ ($T' < 60.4$) (T-021)
class scope	registered shell/shell-union; off-shell = fallback (Sec.5.1)

5. The remaining item, formalised

The closures quantify over the corrected admissible class

$$\mathcal{A}_{\text{adm}} = \left\{ Q \subset \bigcup_{R \in \mathcal{R}} (\Lambda \cap \{|x|^2 = R\}) : \begin{array}{l} |Q| \leq n_{\text{pack}}, \quad Q = -Q, \quad \angle(k_i, k_j) \geq \theta_{\min}, \\ Q \text{ a registered shell/shell-union competitor} \end{array} \right\},$$

on which $T'(Q) \leq 13$. The operator item is the COMPLETENESS of $\mathcal{A}_{\text{adm}}$: that every physically relevant B1 Reading-H competitor is crystalline, shell-supported, separated, and $\leq n_{\text{pack}}$.

5.1 Adversarial-class caveat and fallback bound

The sharp pin $T'(Q) \leq 13$ is a registered pin over $\mathcal{A}_{\text{adm}}$; it is NOT a theorem over arbitrary antipodal lattice subsets. Indeed there exist antipodal lattice sets with

$$N = 40, \quad T'(Q) = 20 > 13,$$

which are lattice-supported but not registered single-shell / shell-union competitors. Any real antipodal competitor obeys the universal geometric fallback

$$T'(Q) \leq \frac{N}{2}$$

(C_t and C_{-t} disjoint for $t \neq 0$; antipodal symmetry maps one onto the other, so $2T' \leq N$). With $N \leq n_{\text{pack}} = 40.7$, hence $N \leq 40$, $T'(Q) \leq 20$, and

$$20 < 20.5 \text{ (RES-1)} < 26.2 \text{ (RES-5)} < 60.4 \text{ (SC-SCOPE)}.$$

Thus off-shell escape competitors falsify the SHARP $T' \leq 13$ pin but do NOT falsify the H-diag, RES-5, or SC-SCOPE thresholds, provided the packing bound $N \leq 40$ and real antipodal symmetry are retained; the cost is thinness (RES-1 $R_{\text{lead}}(20) = 0.974$, $\times 1.026$; SC-SCOPE joint $\times 1.082$). Whether such escape competitors are physically admissible in B1/B2 is the operator question. (Full grading: aadm-exclusion-boundaries v1.1, Boundary 3.)

6. Path to a re-tier (operator decision)

If the operator signs off on $\mathcal{A}_{\text{adm}}$ completeness – the ground state is crystalline AND shell-supported – the analytic residuals are discharged on $\mathcal{A}_{\text{adm}}$ and B1/B2 may be re-examined. The re-tier inherits the thin-certified $R_{\text{max}} = 0.385$, sunset $S = 1.13$, the T-017 pin, and the anchoring; and it must respect the off-shell fallback (RES-1 thins to $\times 1.026$ if off-shell escapes are admitted). The honest ceiling is a NAMED-HYPOTHESIS-REDUCED T6 (H-LAYER $\rightarrow$ shell-supported-crystalline completeness + inherited grades), NOT an unconditional T7. No tier flip is enacted here.

7. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “*v1.0 implied $T' \leq 13$ universally.*” UPHELD and corrected: §5.1 records that $T' \leq 13$ is class-specific and the universal fallback is $T' \leq N/2 \leq 20$.
- (β) “*The thresholds survive the fallback only thinly.*” ACKNOWLEDGED: RES-1 thins to $\times 1.026$ at $T' = 20$. The comfortable margins hold on the registered $T' \leq 13$ class; the shell-supported restriction is what buys them.
- (γ) “*Does this flip B1/B2?*” DISMISSED: no. §6 ceilings the re-tier at a hypothesis-reduced T6. B1/B2 stay T6 on {H-LAYER}.

Quantitative sanity check (mandatory): (D) $\Phi''_{\text{diag}} = 4.03 > 0$, entropy $\geq 4.64 \times 10^{-2}$; (O) $R_{\text{lead}} \leq 0.650 < 1$; (S) $T' \leq 13 < 60.4$, joint $\geq \times 1.095$; *fallback* $T' \leq N/2 \leq 20$, $R_{\text{lead}}(20) = 0.974 < 1$, $20 < 20.5 < 26.2 < 60.4$.

8. Result footer

Result ID:	B2-PROPA-HLAYER / H-LAYER analytic-closure consolidation (v1.1)
Precise statement:	The T-016..T-021 arc closes every ANALYTIC H-LAYER residual class-wide on the registered admissible class $A_{\text{adm}} = \{Q \text{ subset shell-UNION} : Q \leq n_{\text{pack}}, Q = -Q, \text{angle} \geq \text{theta}_{\text{min}}, \text{registered shell/shell-union}\}$, on which $T' \leq 13$. (D) diagonal isotropy strict minimum (T-016); (O) no A-independent Fock exchange (T-019), $\rho \cdot R_{\text{lead}} \leq 0.650 < 1$ (T-018,020); (S) $K_{\text{floor}} \leq T' \leq 13$ (T-017), SC-SCOPE joint $\geq \times 1.095$ $T' < 60.4$ (T-021). The sharp $T' \leq 13$ is CLASS-SPECIFIC; the universal antipodal fallback is $T' \leq N/2 \leq 20$ (escape $N=40, T'=20$ exists), under which $20 < 20.5 < 26.2 < 60.4$ -- the thresholds survive (RES-1 thin $\times 1.026$). Sign-off = crystalline + SHELL-SUPPORTED.
Scope:	Operating endpoint $I = 2e-3$; ROBUSTNESS-MU2 [$\times 0.5, \times 2$]. Registered class = sharp $T' \leq 13$; off-shell = $T' \leq N/2 \leq 20$ fallback.
Dependencies:	isotropy-infimum-core v1.1 (T-016); res5-chi-link-bypass v1.1 (T-017); hdiag-offdiag-gershgorin-threshold v1.1 (T-018); offdiag-exchange-scalar-identification v1.0 (T-019); classwide-secondcumulant-stability v1.0 (T-020); sscope-classwide-endpoint v1.0 (T-021); aadm-exclusion-boundaries v1.1 (Boundary 3 + fallback).
Evidence grade:	CONSOLIDATION (cites executed T-016..T-021, T-023 each 3-6/6).
Reproduction command:	(per-row) see the cited note's script; this note adds no compute.
Falsification gate:	A physical B1 competitor outside A_{adm} (off-shell with $T' > 20$, i.e. $T' > N/2$; or non-crystalline escaping the DR-2 N^{eps} bound) lowering F below the isotropic layer; or a re-opened analytic row.
Tier before / after:	No tier flip. B1/B2 T6 on {H-LAYER}, B5 T5. Analytic residuals class-wide closed on the registered class; off-shell fallback-controlled (RES-1 thin); re-tier is the operator competitor-class decision, ceilinged at hypothesis-reduced T6 (NOT T7).
No-overclaim statement:	Does NOT flip B1/B2 and does NOT claim unconditional closure. The sharp $T' \leq 13$ is class-specific; the universal fallback $T' \leq N/2 \leq 20$ keeps the thresholds alive thinly. The completeness sign-off (crystalline + shell-supported) and the inherited thin grades remain.
Next required action:	OPERATOR: decide whether off-shell / arbitrary multi-radius antipodal lattice subsets are admissible B1/B2 competitors. Shell-supported $\Rightarrow$ comfortable $T' \leq 13$ margins; admitted $\Rightarrow$ RES-1 $\times 1.026$ fallback binds.